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Magnetite Dissolution and Neocrystallization During Cleavage Formation: Paleomagnetic Study of the Martinsburg Formation, Lehigh Gap, Pennsylvania

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Paleomagnetic study of the shale to slate transition in the Ordovician Martinsburg Formation at Lehigh Gap, Pennsylvania reveals that the remanence of the relatively undeformed shales predates cleavage formation, and that the slates were remagnetized during cleavage development. The shales display two components of remanence. Component 1 (C1) was the only magnetization direction in the majority of the shales, and the first-removed component in 10 shale specimens. Component 2 (C2) is observed as a single-component remanence with a low (300-400°C) unblocking temperature range in samples from one shale station and as a second-removed component in 10 shale specimens. The mean directions are (declination/inclination) 347°/-04° (346°/-47°) in situ (tilt-corrected) for C1 and 80°/-63° (135°/-40°) for C2. The pencil slates and well-cleaved slates both display single component remanences. The mean directions are 198°/-12° (201°/26°) for the pencil slates and 199°/24° (225°/54°) for the well-cleaved slates. Using the relationship between cleavage development and remagnetization in the slates, the timing of cleavage development and folding, and the regional geology, we can constrain the relative ages of the remanences as follows: (1) the shale C1 direction is most likely prefolding and represents a primary upper Ordovician direction, (2) the slate directions are younger in age than the shale directions and are probably of Permo-Carboniferous (Alleghenian) age. These results show that remagnetization occurred during development of slaty cleavage in these rocks. We interpret the mechanism for this remagnetization to have been strain-induced dissolution and neocrystallization of magnetite. The additional difference in the characteristic directions of the pencil slates and well-cleaved slates may be attributed to strain reorientation of the pencil slate remanence after new growth of magnetite. The unknown temporal relationship between dissolution and neocrystallization and strain precludes any quantitative attempt to correct remanence for strain in these samples. Our study shows that paleomagnetic studies of deformed sediments must consider the possibility of dissolution and neocrystallization of remanence carriers, in addition to strain-induced rotation.

INTRODUCTION

Paleomagnetic results from folded Paleozoic sedimentary rocks in the Appalachians define portions of the Early Paleozoic apparent polar wander path (APWP) for North America [see *Van der Voo*, 1989]. Recently *Stamatakos and Kodama* [1991a,b] have evaluated the effects of strain on the remanence of two important red bed units, the Carboniferous Mauch Chunk Formation and the Silurian Bloomsburg Formation. They concluded that the remanence directions of these units are deflected from their original, primary direction by strain reorientation of hematite during folding. Thus the paleopoles previously derived from the more strongly deformed parts of these units do not provide an accurate representation of the Earth's field for the depositional age of these units. Similar deflections of remanence carried also by hematite have been observed in folded sediments from the Alps [*Kligfield et al.*, 1983; *Hirt et al.*, 1986] and from Brittany, France [*Cogné et al.*, 1991]. The effects of strain and cleavage development during folding on natural remanence carried by magnetite have not been studied to this extent. With this in mind, we have conducted a detailed paleomagnetic study of the shale to slate transition in the Upper Ordovician Pen Argyl member of the Martinsburg Formation at Lehigh Gap, Pennsylvania (Figure 1).

Regionally, the Lehigh Gap area is in the internal (southeastern) portion of the Valley and Ridge tectonic province of the East Pennsylvania Appalachians and lies on the northern (east-west striking) limb of the Pennsylvania salient (Figure 1, inset). The Pen Argyl member of the Martinsburg Formation is a fine-grained, dark gray mudstone with sandstone interbeds; at the Lehigh Gap the bed thickness ranges from 1 to 10 cm. The contact with the overlying Upper Ordovician-Lower Silurian Shawangunk Formation is an angular unconformity with a difference in dip of 10° between the Shawangunk and the Martinsburg. The unconformity is attributed to regional tilting and erosion resulting from the Ordovician Taconic orogeny [*Stose*, 1930; *Willard and Cleaves*, 1939; *Drake and Epstein*, 1967]. The Lehigh Gap outcrop is on the upright limb of a large fold [*Epstein and Epstein*, 1969]. The strike and dip of the Martinsburg beds here averages 263°/43°N (strike/dip), whereas the cleavage is essentially perpendicular to bedding (Figure 2) with an average strike and dip of 104°/65°S. The structures in the Martinsburg at the Lehigh Gap display a complete transition from relatively undeformed, uncleaved shale, to weakly cleaved pencil slate, to well-cleaved slate over a horizontal (not stratigraphic) distance of 130 m south from the contact with the

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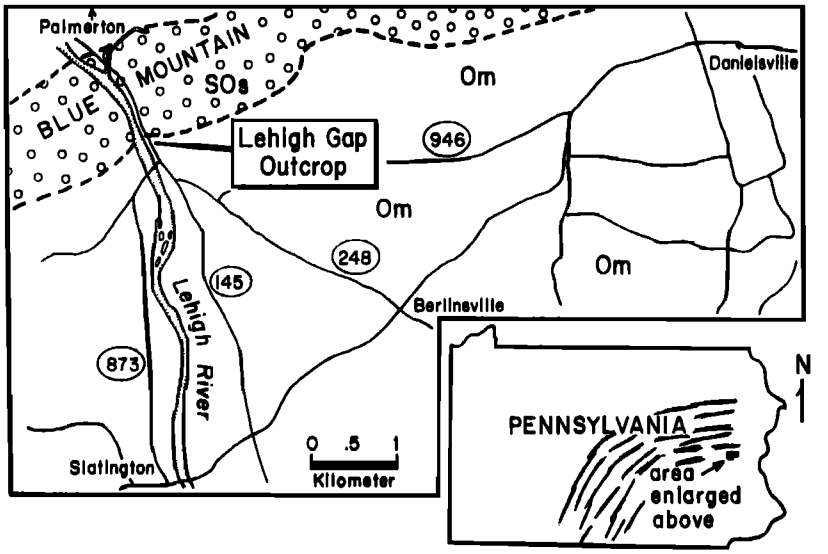


Fig. 1. Geologic map and location of the Lehigh Gap outcrop, with the Martinsburg Formation (Om) and Shawangunk Formation (SOs) denoted. The inset map of Pennsylvania shows the site location on the north limb of the Pennsylvania salient. The salient is defined by regional fold axes trends, which range from NE in the south to E-W in the north [*Engelder and Geiser*, 1980].

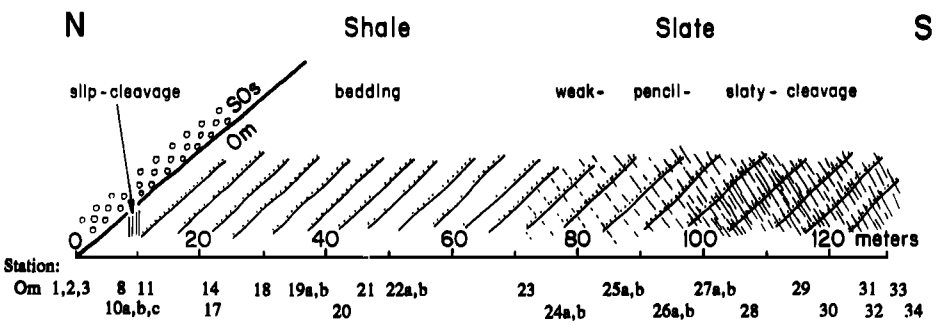


Fig. 2. Schematic diagram of the Lehigh Gap outcrop showing the bedding attitudes and stages of cleavage development. Locations of sample stations (Om1-Om34) and horizontal distance in meters from the Martinsburg-Shawangunk contact are marked at the bottom.

overlying quartzite-conglomerate of the Shawangunk Formation (Figure 2). This transition is the result of a strain shadow beneath the overlying competent quartzite, and the gradual development of cleavage reflects a gradient in strain increasing with distance from the Martinsburg-Shawangunk contact [*Epstein and Epstein*, 1969]. The slaty cleavage is axial planar to, and thus coeval with, the folds that developed during the Permo-Carboniferous Alleghenian deformation event [*Epstein et al.*, 1974]. Microstructural [*Holeywell and Tullis*, 1975; *Woodland*, 1982; *Wright and Platt*, 1982; *Lee et al.*, 1986] and magnetic fabric studies [*Housen and van der Pluijm*, 1990] have shown that the cleavage developed via dissolution and new growth of phyllosilicates. The peak metamorphic conditions in the Martinsburg were subgreenschist to lowest greenschist facies, with a maximum temperature of 150°C [*Wintsch*, 1978] to 300°C [*Epstein et al.*, 1977].

SAMPLING AND METHODOLOGY

Samples were collected at 34 sample stations along the outcrop (Figure 2). The distance from the Martinsburg-Shawangunk contact in Figure 2, and cited in this paper, is measured horizontally. Samples from Om3 were collected with a portable drill, all others were collected as oriented blocks. Several standard 2.5-cm-diameter, 2.2-cm-length specimen cores were prepared from each sample station, yielding a total of 100 specimens. For this study the sample stations are divided into three sample sites: site 1 containing all shale samples (Om1-Om22B, 0-55 m from the contact), site 2 containing all pencil slates (Om23-Om26A, 74-97 m), and site 3 containing all well-cleaved slates (Om26B-Om34, 100-130 m).

The remanence measurements were made using a Sc-T two-axis cryogenic magnetometer. Both incremental thermal (15-20 heating steps) and alternating-field (AF; 18-20 peak field steps) demagnetization treatments were performed using Schonstedt demagnetization equipment. Principal component analysis [*Kirschvink*, 1980] based on orthogonal vector plots [*Zijderveld*, 1967] was used to determine characteristic remanence directions in each sample. The measurement and demagnetization instruments are housed in a magnetic field-free room in the University of Michigan Paleomagnetic Laboratory. Experiments to determine and characterize the magnetic mineralogy of these samples included acquisition of saturation isothermal remanence (sIRM), thermal demagnetization of multi-component IRM, and acquisition of partial anhysteretic remanence (pARM).

Additionally, scanning electron microscopy (SEM) was utilized to identify the magnetic minerals present in these samples. All SEM observations were obtained from polished thin sections. Compositions were determined qualitatively through a Kevex energy dispersive spectrometer attached to the SEM at the University of Michigan Electron Microbeam Analysis Laboratory. Additional information about the behavior of the magnetic minerals in these rocks is provided by the study of ARM anisotropy (ARMA) of *Housen and van der Pluijm* [1991].

RESULTS

All the samples have low NRM magnitudes ranging from 0.3 to 0.05 mA/m. Owing to alteration during heating and because of weaker remanence intensities, one third of the samples did not yield stable demagnetization behavior and were excluded from further consideration.

Site 1: Shale

Most of the shale samples contained essentially a single-component remanence direction (Figures 3a and 3b). This direction was completely removed by thermal demagnetization at temperatures from 540° to 580°C and by AF demagnetization to 100 mT. This direction will be referred to as shale component 1 (C1). Both the unblocking temperatures and coercivities are distributed rather than discrete, indicating a distribution of grain sizes and/or compositions of the carrier of this remanence. A site mean direction was calculated for the 42 specimens (out of 60 total) which displayed a stable C1 direction. The mean C1 direction in geographic coordinates is (declination/inclination) 347°/-4°, κ= 16, α_{95}= 6°; in stratigraphic coordinates the mean is 346°/-47°, κ= 18, α_{95}= 6°.

Another remanence direction was observed in 15 shale specimens, either as a second-removed magnetization component

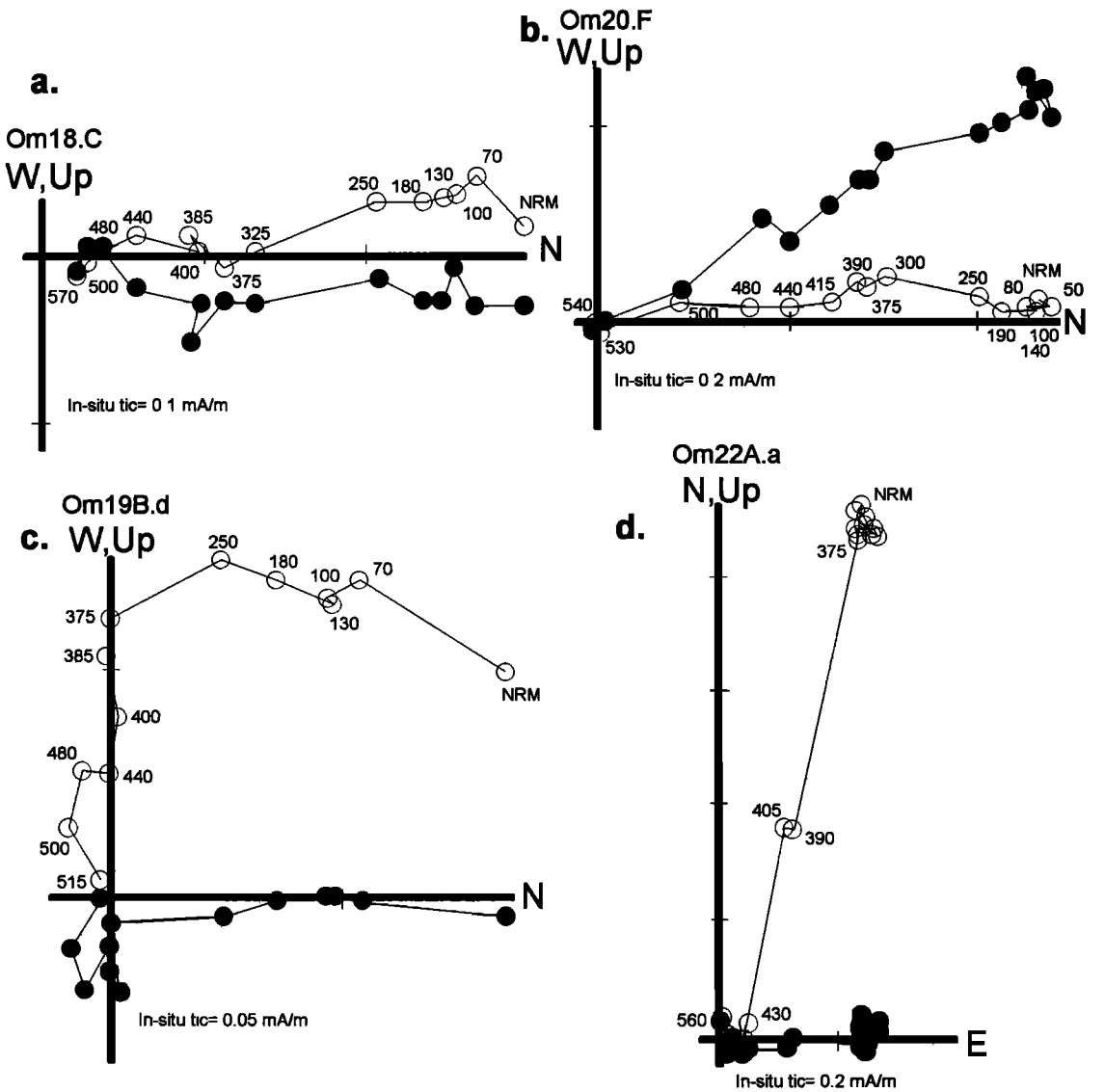


Fig. 3. Orthogonal demagnetization [*Zijderveld*, 1967] diagrams showing characteristic remanence directions that were observed during step-wise thermal demagnetization of the shale samples. Solid (open) symbols are projections on the horizontal (vertical) plane. All figures in geographic coordinates. Remanence intensities are given in mA/m. Temperatures (°C) for each step are indicated. Figures 3a and 3b show single-component shale C1 directions. Figure 3c shows a specimen with both shale C1 and C2 directions; Figure 3d shows a specimen with a single-component C2 direction.

(Figure 3c) or as a single-component remanence (Figure 3d). This direction will be referred to as shale component 2 (C2). In specimens with both shale C1 and C2, C2 is removed between 375° to 580°C (Figure 3c). In most cases the C2 direction had either a slightly curved trajectory on the demagnetization plot or was defined by only three points. In those cases it is difficult to confidently determine the characteristic C2 direction using either least-squares fit of the vector or great circle pole techniques of principal component analysis. The C2 directions from these specimens may be contaminated by incompletely removed C1 directions. The specimens which display the C2 direction as a single component of magnetization are all from sample station Om22A. In these specimens the C2 direction is removed at a well-defined unblocking interval between 375° and 430°C (Figure 3d). Alternating field demagnetization was ineffective on the Om22A specimens. The Om22A specimens provided characteristic C2 directions which are nearly identical to the C2 directions obtained from the other shale specimens, thereby arguing against significant contamination of the C2 direction in the other specimens. The mean C2 direction in geographic coordinates is 80°/-63°, κ= 22, α_{95}= 8°; in stratigraphic coordinates the mean direction is 135°/-40°, κ= 22, α_{95}= 8°.

Site 2: Pencil Slate

The pencil slate specimens all display a single component of magnetization, which is progressively removed by thermal treatment between 250° and 580°C and by AF treatment up to 100 mT (Figure 4). Like those of the shales, the unblocking temperatures are distributed, indicating a variable grainsize and/or composition of the remanence carrier. The 10 pencil slate specimens define the characteristic pencil slate direction as 198°/-12°, κ= 10, α_{95}= 16° in geographic coordinates, and 201°/26°, κ= 10, α_{95}= 16° in stratigraphic coordinates.

Site 3: Well-cleaved Slate

The well-cleaved slates also display only single-component magnetization directions (Figure 5). The unblocking temperatures and coercivities for the slate specimens are also similar to those of the pencil slates. The mean characteristic slate direction, from principal component analysis of 15 slate specimens, is 199°/24°, κ= 15, α_{95}= 11° in geographic coordinates, and 225°/54°, κ= 15, α_{95}= 11° in stratigraphic coordinates.

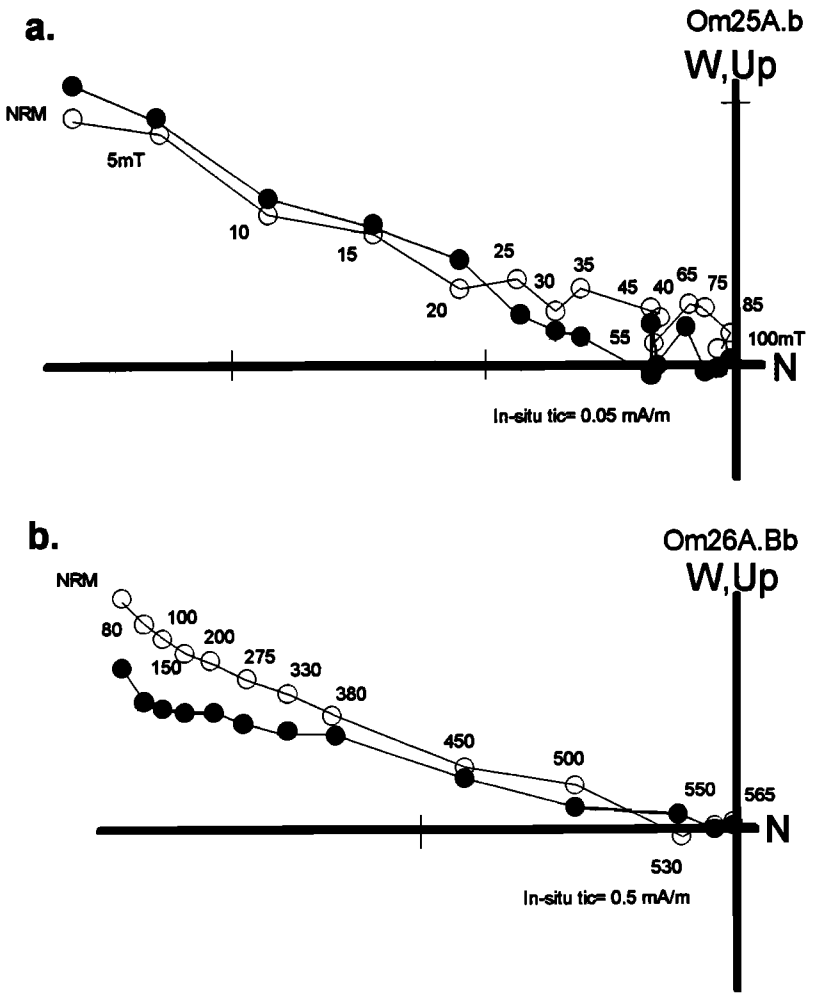


Fig. 4. Orthogonal demagnetization diagrams showing characteristic remanence directions in the pencil slates. (a) A result obtained from thermal treatment. (b) A result obtained via AF demagnetization, with the peak AF given in milliteslas. Symbols and projection as in Figure 3.

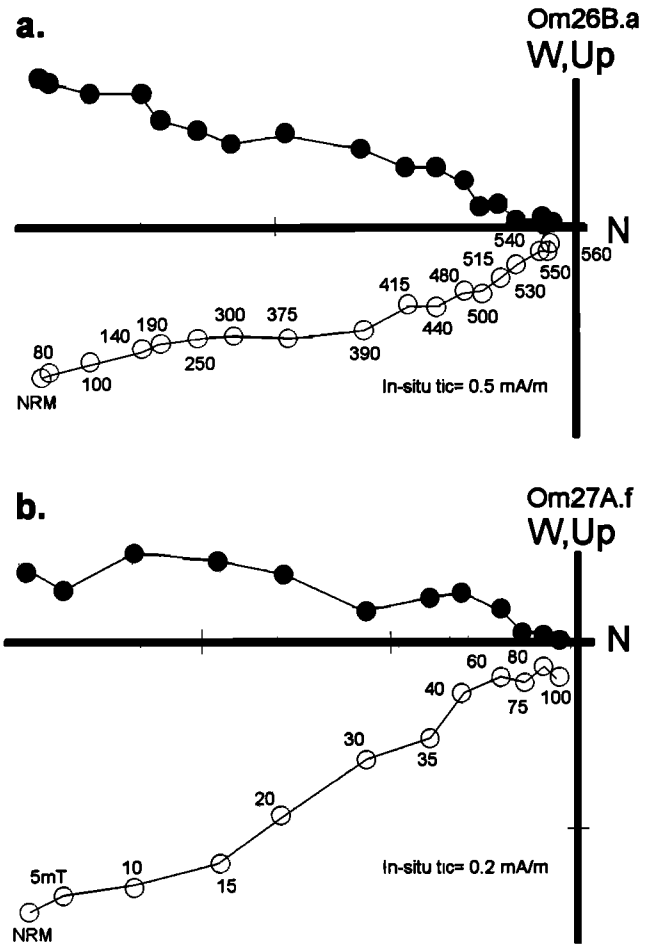


Fig. 5. Orthogonal demagnetization diagrams showing characteristic remanence directions that were observed in well-cleaved slates. (a) A result from thermal treatment. (b) From AF treatment. Projections and symbols as in Figure 3.

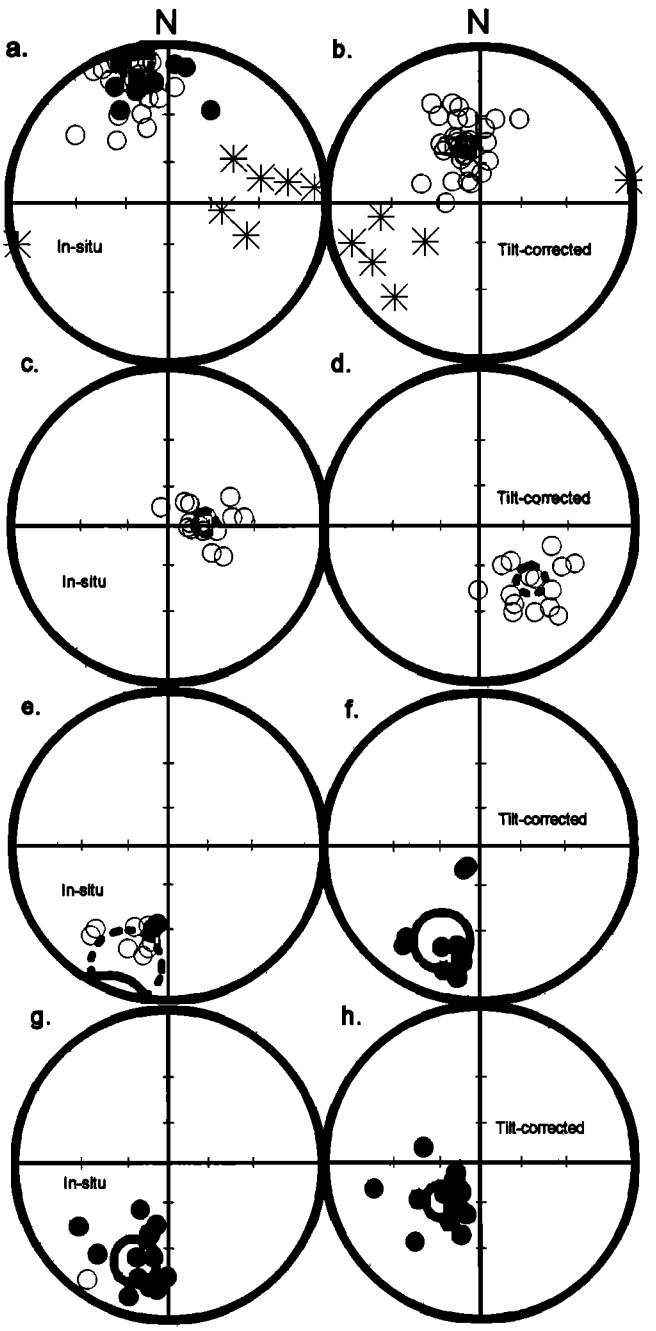


Fig. 6. Stereographic (equal angle) projections of the characteristic directions observed in the Martinsburg Formation samples. Solid symbols are lower hemisphere and open symbols are upper hemisphere projections. Asterisks represent poles to great circles. Each symbol is the result from a single specimen. The mean direction for each component is also plotted with its corresponding circle of 95% confidence. For the shale C1 direction (Figures 6a and 6b), great circle analyses were used for seven of the specimens in which both shale components were removed simultaneously. The shale C2 directions are shown in Figures 6c and 6d. The pencil slate directions are plotted in Figures 6e and 6f; and the well-cleaved slate directions are plotted in Figures 6g and 6h. Mean directions and their statistics are given in Table 1.

SUMMARY OF PALEOMAGNETIC DATA

Using the mean directions (Figure 6) and the location of the site, paleomagnetic poles are calculated for the shale C1 and C2 directions, the pencil slate direction, and the well-cleaved slate direction (Table 1). Because the dip and strike of the beds throughout the outcrop are fairly uniform, a fold test is not possible. Lacking a traditional paleomagnetic field test to constrain the age of magnetization in the Martinsburg Formation, we include paleopoles using both geographic and stratigraphic coordinates in Table 1. We will attempt to identify the geologically meaningful paleopoles by considering the following: (1) the poles should match other North American poles of Ordovician depositional age or poles with ages of probable remagnetization events and (2) possible structural complexities such as rigid body rotation associated with Pennsylvania salient development, strain alteration of remanence, and synfolding remagnetization.

MAGNETIC MINERALOGY

The sIRM results for both shale and slate (Figure 7) samples show saturation between 0.2 and 0.5 T, which is consistent with the presence of either magnetite or coarse-grained pyrrhotite [*Dekkers*, 1988]. Using the technique of *Lowrie* [1990] to differentiate between IRMs carried by magnetite or pyrrhotite, we imparted multicomponent IRMs (mIRM) to several samples.

TABLE 1. Summary of Paleomagnetic Data From the Martinsburg Formation

	N	D/I	κ	α_{95}	Pole
Shale C1	42				
In situ		347°/-4°	16	6°	45°N, 124°E
Tilt corrected		346°/-47°	16	6°	20°N, 119°E*
Shale C2	15				
In situ		080°/-63°	22	8°	21°N, 237°E
Tilt corrected		135°/-40°	22	8°	50°N, 178°E
Pencil slate	10				
In situ		198°/-12°	10	16°	52°N, 75°E
Tilt corrected		201°/26°	10	16°	33°N, 80°E
Well-cleaved slate	15				
In situ		199°/24°	15	11°	34°N, 82°E
Tilt corrected		225°/54°	15	11°	5°N, 69°E

N is the number of specimens, D/I are the declination/inclination of the mean direction, α_{95} and κ are the precision statistics for the mean direction.
*Results after correction for 23° clockwise rotation of the Pennsylvania salient are 13°N, 138°E.

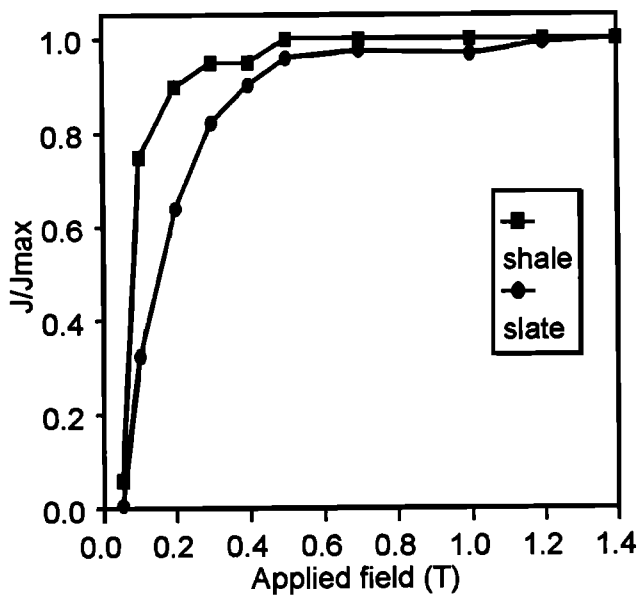


Fig. 7. Saturation IRM experiment results for shale (circle) and slate (square) samples. The vertical axis is J/J_{max}, the horizontal axis is applied field (T). Both plots show saturation of IRM between 0.3 T and 0.5 T, which is indicative of either fine-grained magnetite or pyrrhotite.

Fields of 1.4, 0.4, and 0.1 T were successively applied in orthogonal directions. Thermal demagnetization of the multicomponent IRM allows the use of both the coercivity and thermal unblocking characteristics of magnetic minerals in the identification of the remanence carriers in a sample. In the shales (Figure 8a) the majority of the mIRM is carried by the low-coercivity component, which thermally unblocks at 580°C. This indicates that the dominant magnetic mineral in the shales is magnetite. Analysis of mIRM of a shale sample from Om22A, which carries the C2 as a single component of magnetization, shows that all three (high, intermediate, and low) coercivity components unblock significantly between 300° and 400°C (Figure 8b). This is similar to the observed unblocking temperatures of the C2 direction in the natural remanence of these samples. Possible carriers of this magnetization are pyrrhotite, greigite, titanomagnetite, or maghemite. Greigite is chemically unstable and has not been reported in rocks older than Miocene in age. Titanomagnetite has coercivities less than 0.3 T and thus may carry the 0.1 T and the 0.4 T components of the mIRM, but not the higher-coercivity component. Maghemite unblocks between 300° and 400°C and could carry all three mIRM components. At higher temperatures, however, maghemite converts to hematite, which would produce magnetization components with unblocking temperatures up to 680°C. Although such high unblocking temperature components are not observed, maghemite may not be ruled out as a carrier of the C2 direction. Another likely carrier of this magnetization is pyrrhotite, whose coercivity ranges from 0.3 to 0.7 T [*Dekkers*, 1988]. Although the Curie temperature of pyrrhotite is reported to be about 310°C [*Piper*, 1987], *Dekkers* [1988] has reported that it may in some cases actually be higher. The T_{ub} range of 300-400°C observed in all three IRM components of Figure 8b may also therefore be attributable to a remanence carried by pyrrhotite. Approximately 30% of the initial mIRM in the Om22A samples is removed between 400° and 580°C, indicating that magnetite is also an important remanence carrier in these samples. The slate samples display mIRMs with a dominant remanence that has low coercivity and is gradually removed by thermal treatment between 200° and 580°C, which indicates that the slate remanence is carried by magnetite (Figure 8c). A low unblocking temperature, high coercivity component indicates the presence of goethite. Thus the sIRM and mIRM experiments indicate that the principal remanence carrier in the shales (excepting shale Om22A), pencil slates, and slates is magnetite,

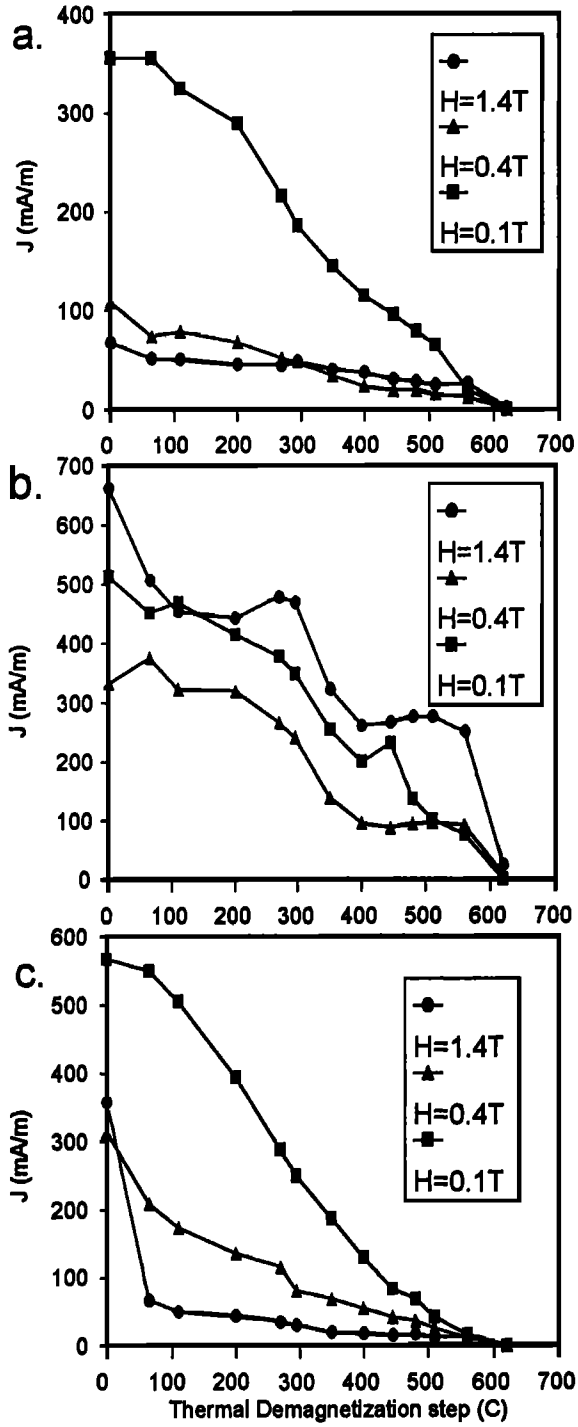


Fig. 8. Thermal demagnetization of multi-component IRM, using the method of *Lowrie* [1990]. The vertical axis is remanence intensity (mA/m), the horizontal axis is thermal demagnetization temperature. The three orthogonal components are 1.4 T (hard), 0.4 T (medium), 0.1 T (soft), and are denoted by open squares, solid triangles, and solid squares, respectively. (a) Shale specimen showing behavior dominated by a soft component which is removed by 580°C. (b) Shale Om22A specimen showing significant contributions from the hard, medium, and soft components. These components unblock in two intervals, 300° to 380°C and 580° to 610°C. (c) slate specimen, with a hard low T_{ub} and a soft component which is removed by 580°C.

whereas the shale samples of Om22A probably also contain significant amounts of pyrrhotite and/or maghemite.

Measurements of partial ARM (pARM) were conducted to determine the grainsize distribution of the magnetite in the Martinsburg samples using the method of *Jackson et al.* [1989]. Results from a typical shale, pencil slate, and well-cleaved slate show coercivity spectra with peak pARM intensities between 15 and 35 mT (Figure 9). This indicates that the magnetite grainsize distribution across the shale to slate transition is fairly uniform, with an average magnetite grain size of 2-5 μm.

Scanning electron microscopy (SEM) was conducted to further characterize the magnetic minerals in the shales and slates. This examination revealed three morphological types of iron-oxide grains in the Martinsburg samples: blocky, pseudoframboidal, and spherical. Block-shaped iron-oxide grains were found in both slate samples (Figure 10a) and shale samples (Figure 10b). Most of the iron-oxides observed via SEM in both shales and slates are blocky and range in size from 2 to 10 μm. Iron-oxide pseudoframboids were also observed in both shales (Figure 10c) and slates (Figure 10d). Spherical iron-oxide grains were also, less commonly, found in both shales and slates (Figure 10e, with an adjacent blocky magnetite). These spherical grains range in size from 5 to 10 μm. We note that they are not biogenic in origin, as biogenic magnetite is thought to be exclusively single-domain (<1 μm) in size [*Stolz et al.*, 1990]. Both the spherical and pseudoframboidal iron-oxides have morphologies which are consistent with a secondary origin. The morphology of the blocky

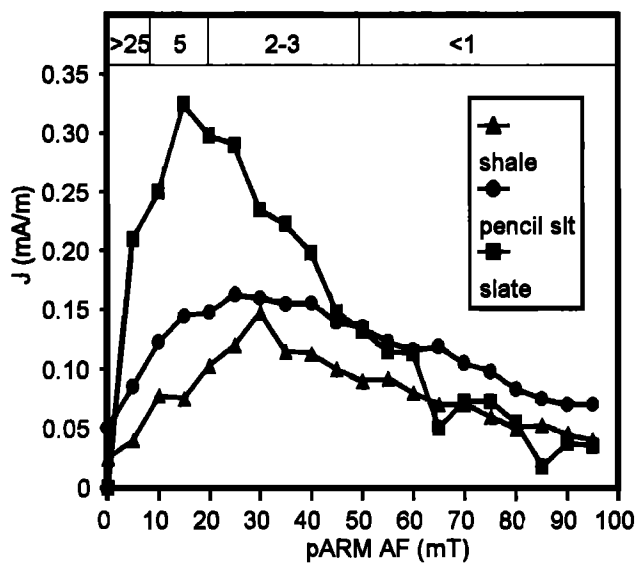


Fig. 9. Partial ARM (pARM) acquisition curves for representative shale (triangle), pencil slate (circle), and well-cleaved slate (square) samples. The pARMs were generated with a steady field of 0.1mT applied over a 5-mT-wide alternating field (AF) window in 20 steps centered on AF values from 0 to 100 mT. Magnetite grain sizes (μm) for given coercivities are provided on the top of the figure. Both shale and slate samples display pARM peaks between 15 and 30 mT, which indicates that the magnetite has an average grain size of 2-5 μm.

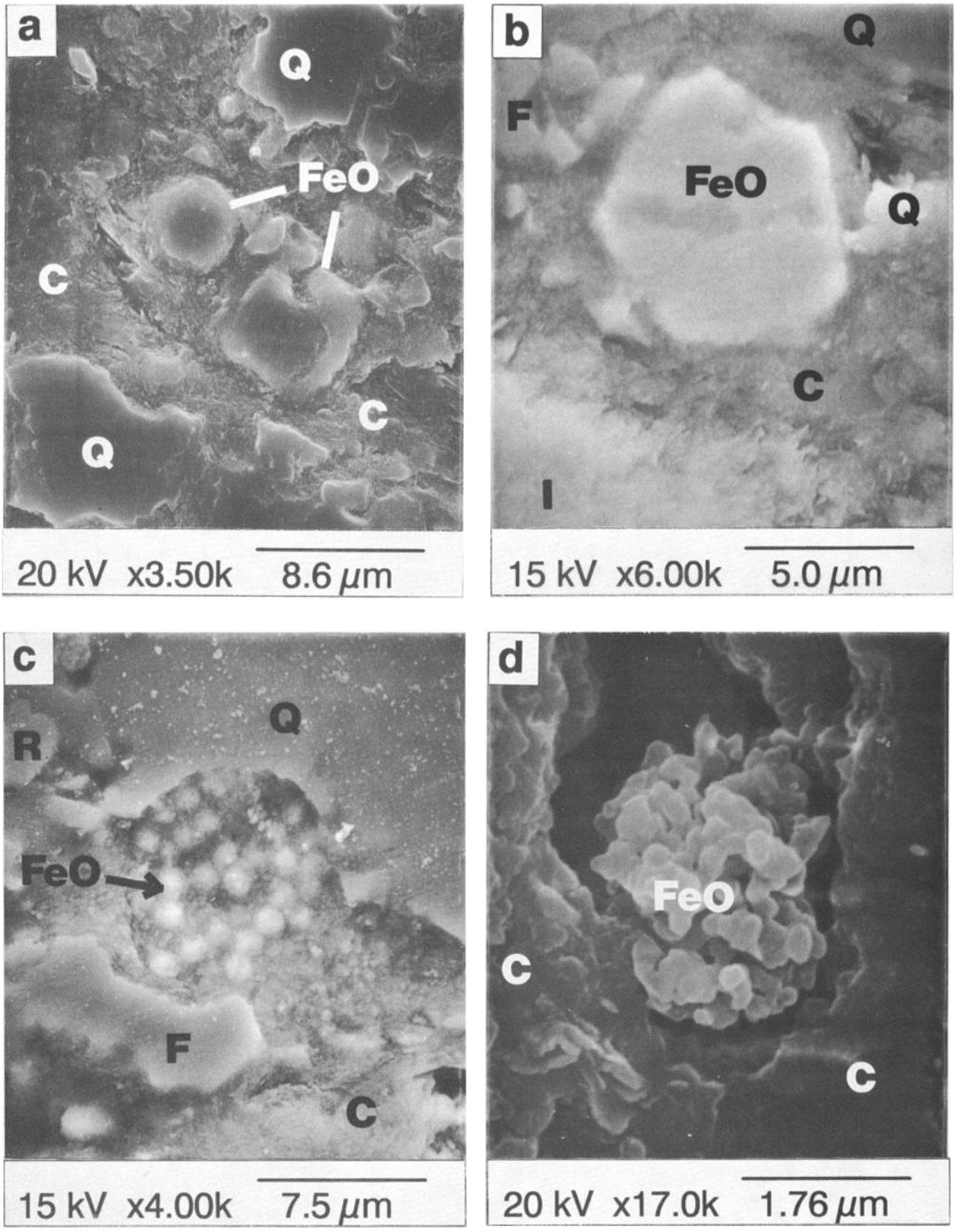


Fig. 10. SEM secondary electron images of blocky iron oxides (magnetite?) from (a) slate and (b) shale specimens. (c) and (d) SEM secondary electron images of iron oxide (magnetite?) pseudoframboids in slate specimens. In each figure the iron oxides are subspherical aggregates of 0.1-0.7 μm cubes. (e) SEM secondary electron image of spherical iron oxide (magnetite?) (top center) with a blocky iron oxide. A similar sphere is seen in Figure 10b. (f) SEM secondary electron image of pyrite framboids observed in sample Om22A. Scale bar, magnification, and accelerating voltage are shown on the bottom of each figure. For this figure, mineral abbreviations are FeO, iron oxide; Q, quartz; I, illite; F, feldspar; R, rutile; C, chlorite; Pyr, pyrite.

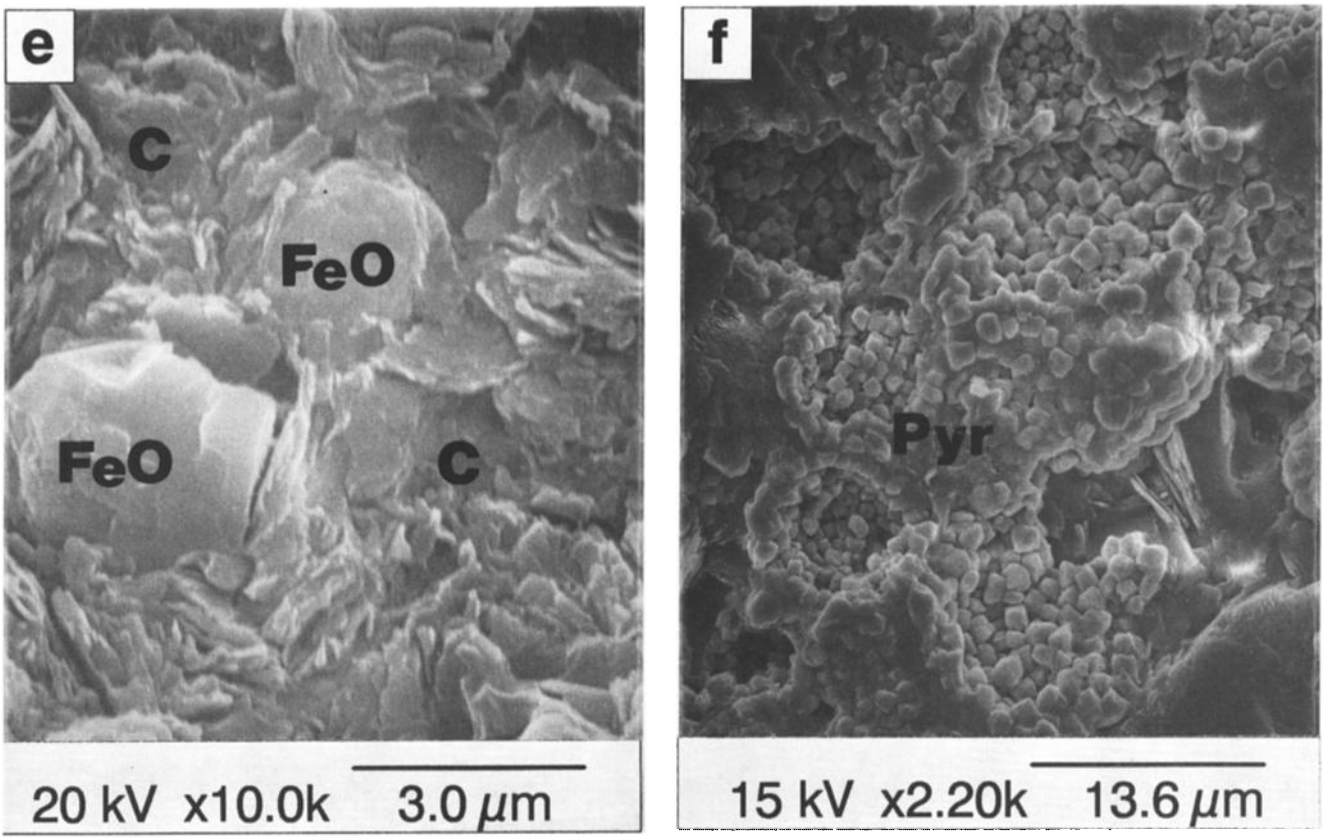


Fig. 10. (continued)

iron-oxide is consistent with either a detrital origin or a secondary diagenetic or metamorphic origin. Iron is the only cation observed in all three types of the iron-oxides. Although we have no crystallographic data we infer that these iron-oxides are magnetite based on our rock magnetic work that indicates magnetite is the principal magnetic mineral in these rocks and that hematite is not present. Additionally, magnetite with morphologies similar to those we found in the Martinsburg has been previously observed and characterized in great detail by SEM and TEM studies of *Suk et al.* [1990] and *Suk* [1991] in Paleozoic carbonates of eastern North America (e.g., the Trenton and Onondaga Limestones). These studies found that pseudoframboidal magnetite has replaced framboidal pyrite, retaining the morphology of the replaced pyrite. The size of the individual magnetite grains which comprise the pseudoframboids in the work of *Suk et al.* [1990] and *Suk* [1991] is 0.4-1.0 μm, which is similar to the individual grains in the pseudoframboids we have found. Provided that these grains do not have significant magnetic interactions, they should be capable of carrying a stable remanence for long periods of time. The pseudoframboidal magnetites observed by *Suk et al.* [1990] display both complete and partial replacement of the iron sulfides by magnetite. The qualitative compositions of the pseudoframboids and spheres in the Martinsburg Formation show only iron oxide compositions, indicating complete replacement of the iron sulfides by magnetite. Samples from station Om22A are an exception; they contain abundant pyrite framboids that were not replaced by iron oxides. The Om22A samples were the only samples in which unreplaced pyrite framboids (Figure 10f) as well as pyrite cubes were observed.

All three types of magnetite in the Martinsburg samples are possible carriers of the observed remanence directions. The distributed unblocking temperatures during thermal demagnetization suggest a spectrum of grainsizes as carriers of the remanence. Such a spectrum of grainsizes is best exhibited in the blocky magnetite (Figures 10a and 10b). This observation and the relative abundance of this magnetite phase in both the shale and slate samples suggest that the blocky magnetite may be the principal remanence carrier in the shales, pencil slates, and well-cleaved slates. Additionally, the size of the blocky magnetite is consistent with the size estimate obtained from ARMA study by *Housen and van der Pluijm* [1991]. We cannot, however, rule out significant contributions to the remanence from the spherical magnetite and/or the magnetite pseudoframboids. Additionally, since all three morphological types of magnetite are found in the shales, pencil slates, and slates, we cannot uniquely propose any single remagnetization process solely from the petrologic results.

DISCUSSION

The lack of traditional paleomagnetic field tests prevents us from making a unique determination of the ages of the remanence directions observed in the Martinsburg Formation shales and slates. The slate remanence directions are, however, clearly different from those of the shales. There are essentially three ways to produce this difference: (1) thermal remanent remagnetization of the slates, (2) strain reorientation of the shale direction to the slate direction during deformation, and (3) chemical remanent remagnetization of the slates via dissolution

and neocrystallization of magnetite. A thermoremanence acquired in the slates, but not the shales, in a single 130-m outcrop is not reasonably possible, so we reject this option. The possibility of strain deflections of remanence during deformation was first raised by *Graham* [1949], and it has since been explored using numerical models [*van der Pluijm,* 1987; *Kodama,* 1988], experimental studies [*Kodama and Cox,* 1978; *Borradaile and Mothersill,* 1991; *Borradaile,* 1991, *Kodama and Goldstein,* 1991], and studies of natural examples [*Kligfield et al.,* 1983; *Cogné and Perroud,* 1985; *Hirt et al.,* 1986; *Cogné et al.,* 1991; *Stamatakos and Kodama,* 1991a,b]. All of these studies concluded that remanence is deflected away from the direction of principal shortening *(Z)*, toward the plane of flattening. In cleaved rocks this implies that remanence will be deflected toward the cleavage plane, assuming that the cleavage plane corresponds to the *XY* plane of the finite strain ellipsoid [*Ramsay and Huber,* 1983]. In our case, such a geometry requires a rotation of about 140° to move the shale mean direction to the pencil slate mean direction, with about 100° of this rotation occurring past the cleavage plane. This amount of rotation would require an unrealistically large amount of shear strain, so we reject the possibility that the slate directions represent a strain-rotated shale direction. Additionally, if the shale C1 direction was reoriented during deformation, it would show a gradual change from the shale mean direction to the slate directions. The shale directions do not display this trend, providing further evidence against strain modification of the shale C1 directions. We therefore regard dissolution and neocrystallization of magnetite during cleavage formation as the most likely explanation of the difference between the remanence directions found in the shales and in the slates. We will explore this possibility in greater detail below by examining the relationship between magnetite orientation and slaty cleavage development.

Magnetite and Cleavage

Housen and van der Pluijm [1991] examined the ARM anisotropy (ARMA) of these samples and found that the preferred orientation of magnetite was bedding parallel in the shales and cleavage parallel in the well-cleaved slates. They concluded that this change in preferred orientation was most likely accomplished via dissolution and new growth of magnetite during cleavage formation. This conclusion was supported by experimental and numerical modeling of the Martinsburg ARMA which demonstrated that the measured ARMA tensor in the pencil slates is a composite of two orthogonal magnetite fabrics, one bedding-parallel, the other cleavage-parallel [*Housen et al.,* 1993]. The preferred orientation of magnetite in the shale is consistent with a depositional or early diagenetic origin for this magnetite fabric. Similar depositional magnetite fabrics have been reported from studies of other shale units [e.g. *Jackson et al.,* 1989]. In the well-cleaved slates, the magnetite has a preferred orientation that is parallel to the cleavage. This indicates that the change in magnetite orientation from bedding-parallel to cleavage-parallel occurred in response to the same deformation conditions that produced the cleavage. The bedding-parallel magnetite orientations in the shales indicate that the magnetite in the shales was relatively unaffected by this same cleavage-producing deformation. These findings are consistent with our paleomagnetic results for the slates, which we interpret to be remagnetized, and for the shales, which indicate no modification of the remanence due to deformation. We therefore conclude that the characteristic remanence of all the slates must be younger than that of the shales, which acquired remanence before the formation of slaty cleavage. We interpret the remagnetization of the slate to be the result of magnetite dissolution and neocrystallization in response to strain during cleavage formation. The shale has also escaped the otherwise pervasive Lower Permian, reversed-polarity (Kiaman) remagnetization that is present in nearly all magnetite-bearing and many of the hematitic Paleozoic sediments in the Appalachians [*Elmore and McCabe,* 1991].

Constraints From Structural Geology

Studies of slaty cleavage in folded sediments have long noted [*Harker,* 1885, *Siddans,* 1972] the nearly parallel orientation of the axial plane of folds and slaty cleavage; such cleavage is termed axial-planar cleavage. In their discussion and review of slaty cleavage development, *Ramsay and Huber* [1983] emphasize that the geometric relationships between axial-planar cleavage and folds require that the cleavage must be developed under the same strain conditions that produced the folds in a given area. This means that axial-planar cleavage development and folding are contemporaneous expressions of the same deformation regime. Most studies of the local and regional geology, notably the geologic map and report by *Epstein et al.* [1974], reach the same conclusion; the cleavage in the Ordovician Martinsburg Formation and younger Silurian-Carboniferous units in the area is subparallel to the axial planes of the regional folds and is therefore Upper Carboniferous-Lower Permian (i.e., Alleghenian) in age. Other workers have, however, suggested that at least some of the observed cleavage in this area was formed shortly prior to folding primarily by layer-parallel shortening, but this is still of Alleghenian age [*Geiser,* 1988]. The Alleghenian age of the cleavage is also supported by recent $^{40}Ar/^{39}Ar$ results [*Wintsch and Kunk,* 1992] for micas in the Martinsburg slates from the Lehigh Gap outcrop.

Returning to our paleomagnetic results, we concluded in the previous section that the characteristic remanence in the slates is contemporaneous with cleavage development and that the shale characteristic direction predates the cleavage. Because the cleavage formed either during or shortly prior to folding, the shale remanence direction must predate the folding. If the characteristic directions in the slates were acquired during cleavage formation, strain-induced reorientation of the remanence should be considered. Evaluation of the effects of strain on remanence in natural rocks is done by seeking a correlation between measured strain values and differences between observed remanence directions in undeformed and deformed rocks. Determination of the possible effects of strain on the remanence of the Martinsburg Formation slates is hampered by a lack of strain markers. More importantly, dissolution and neocrystallization of both the matrix phyllosilicates and magnetite preclude any meaningful estimate of finite strain by either AMS or ARMA. We do know, however, that strain

increases away from the Martinsburg-Shawangunk contact which allows us to use the distance from the contact as a qualitative measure of the strain difference between the pencil slates and the well-cleaved slates and to determine if there is any relationship between this strain gradient and the remanence directions. Figure 11a is a plot of distance from the Martinsburg-Shawangunk contact versus the remanence inclinations for the pencil slate and well-cleaved slate specimens. Although the relationship is not well defined, the inclination tends to steepen towards the plane of slaty cleavage with distance. Figure 11b is a sketch of the relationship between the pencil slate and well-cleaved slate directions, the cleavage *(XY)* plane, and the pole to cleavage *(Z)*. The geometry is consistent with the deflection of the remanence in the slates toward the cleavage plane under progressive strain. The feasibility of this interpretation can be tested by calculating the amount of strain necessary to deflect the pencil slate remanence direction to the slate remanence direction. The amount of shortening strain *(Z)* required to achieve the angular difference between the pencil slate mean direction and the well-cleaved slate mean direction can be calculated using the relationship

$$(\tan \theta')/(\tan \theta) = Z/X \qquad (1)$$

where θ and θ' are the angles between the cleavage plane and the pencil slate and well-cleaved slate mean directions, respectively (Figure 11b), and where X, Y and Z are the axes of the finite strain ellipsoid. *Wintsch et al.* [1991] determined that cleavage formation in the Martinsburg Formation at Lehigh Gap was a constant volume process; thus $X*Y*Z=1$. Adding the assumption that the strain associated with cleavage formation approximates uniaxial flattening $(X=Y>Z)$ further simplifies the volume relationship to $X^2*Z=1$. Thus, for the three-dimensional case,

$$(\tan \theta')/(\tan \theta) = Z^{1.5}. \qquad (2)$$

The value of shortening (e) is determined by $e = 1-Z$. The values of θ and θ', the angles between the pencil slate and well-cleaved slate directions and the cleavage *(XY)* plane, are 78° and 42°, respectively. Using these values in (2) gives a shortening strain of $e = 0.66$ (66% shortening). If, on the other hand, we assume constant volume, plane strain conditions $(X>Y=1>Z$ and $X=1/Z)$, the value of Z is 0.44, which represents a finite shortening strain of 56%. Shortening strains typical for slates range from 40 to 60% [*Siddans*, 1972], so the difference in remanence directions between the pencil slates and the well-cleaved slates may be explained by strain-induced reorientation. Because a unique quantitative relationship between strain and dissolution and neocrystallization does not exist, a strain correction on the Martinsburg slate remanence, such as proposed by *Cogné and Perroud* [1985] is not possible. Additionally, bedding-parallel magnetite in the pencil slates is suggested by the composite nature of the pencil-slate ARMA fabrics [*Housen et al.*, 1993], which suggests the possibility that the pencil slate remanence may be a composite direction. Again, a quantitative correction of the pencil slate direction for this effect is not possible. Because of these uncertainties, we do not consider either the pencil slate or the well-cleaved slate directions reliable for further paleomagnetic analysis.

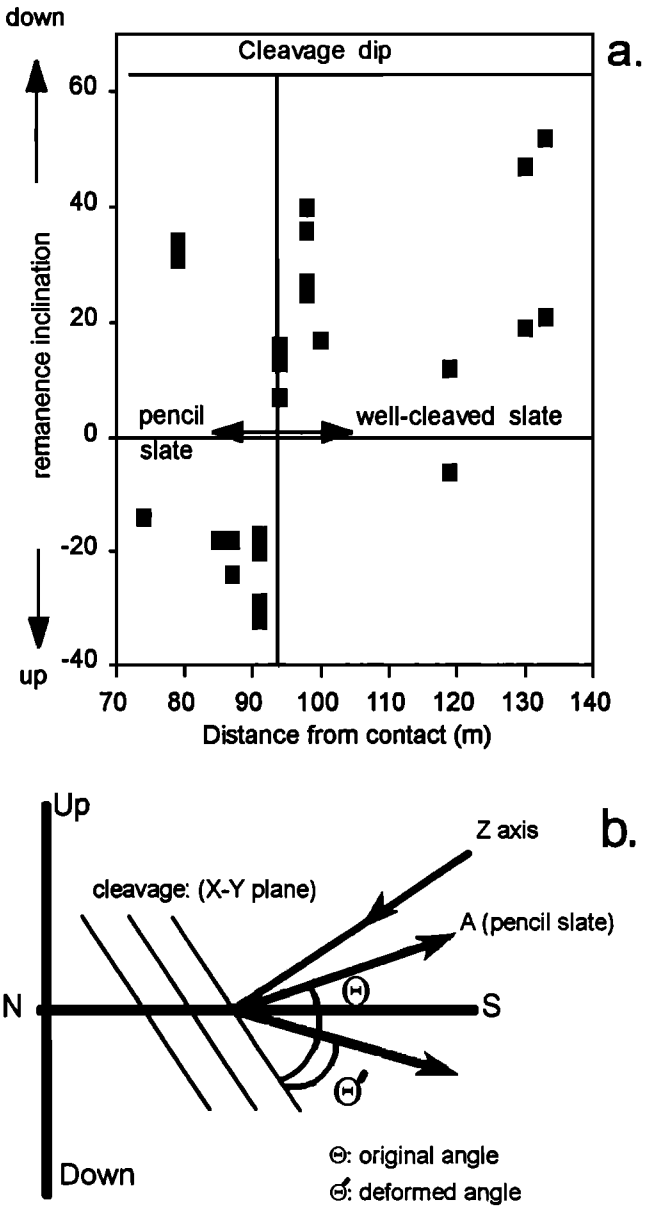


Fig. 11. Relationship between the pencil slate and well-cleaved slate remanence directions and distance from the Martinsburg-Shawangunk contact. The progressive development of cleavage away from the contact indicates increasing strain with distance. (a) Plot of remanence inclination versus distance from the contact. Each symbol represents a single specimen; the pencil slates and the well-cleaved slates are separated by the vertical dotted line. The mean inclination of the pencil slates is marked by point A, the well-cleaved slate mean by B. The plot shows an imperfect trend from shallow upward inclinations to intermediate downward inclinations with distance. (b) Schematic diagram of the slate directions and cleavage orientation. The cleavage plane corresponds to the *XY* plane; the pole to cleavage is the *Z* axis. Strain reorientation of any remanence rotates the remanence from the *Z* axis towards the *XY* plane. A and B are the pencil slate and well-cleaved slate directions as in Figure 11a. The angles of the pencil slate (θ) and well-cleaved slate (θ') directions from the cleavage plane are denoted. This diagram shows that the steepening of the remanence inclination is consistent with strain reorientation of the pencil slate direction to produce the well-cleaved slate direction.

Comparison With the North American APWP

We have concluded from magnetite remanence and orientation and from structural geology that the characteristic remanence of the shale is prefolding in age. An independent test of this conclusion can be made by comparing the poles calculated from the characteristic directions of the shale and the reference North American apparent polar wander path (APWP) for the Cambrian-Jurassic, generated from the global paleomagnetic data base using all poles from the stable craton with $Q \geq 3$ [*Van der Voo*, 1990] (Figure 12). The tilt-corrected shale C2 pole lies far from any portion of the APWP. This direction is best defined by the Om22A samples in which the remanence is probably carried by pyrrhotite. On this basis we exclude the shale C2 direction from further consideration. The tilt-corrected shale C1 pole falls near the lower-middle Silurian and lower Devonian portions of the APWP, which would indicate a possible Silurian to lower Devonian remagnetization in the shales. This remagnetization could represent the conversion of pyrite to secondary magnetite if this process is not an early diagenetic phenomenon. There is, however, no tectonic explanation for a remagnetization of this age, as neither deformation nor metamorphism occurred at this location during this time. Instead, we examine a further structural correction by appealing to the oroclinal bending hypothesis for the Pennsylvania salient, most recently put forth by *Kent* [1988]. The Pennsylvania salient is defined by changes in regional fold axis trends and is a structural feature developed during the Alleghenian orogeny. The angular difference between the southern and northern limbs of the salient is approximately 50°. *Van der Voo* [1990] has argued that the majority of this difference was accomplished via clockwise rotation of the northern limb of the salient while the southern limb was fixed, based on concordances between Lower Paleozoic poles from the southern limb of the salient and poles of similar age from the stable North American craton (e.g., the Silurian Rose Hill and Wabash poles, respectively). The Lehigh Gap is on the northern limb of the orocline so a counter-clockwise rotation of up to 50° may be called for. However, clockwise rotation of the northern limb of the Pennsylvania salient based on paleomagnetic data is 23° ± 12° [*Kent* 1988]. Using this value moves the shale C1 pole to the upper Ordovician part of the APWP (Figure 12). The shale C1 pole is then consistent with the upper Ordovician depositional-early diagenetic age of the Martinsburg Formation.

ALTERNATIVE INTERPRETATIONS

The preceding discussion presents our preferred interpretation of the paleomagnetic data of the Martinsburg Formation. As

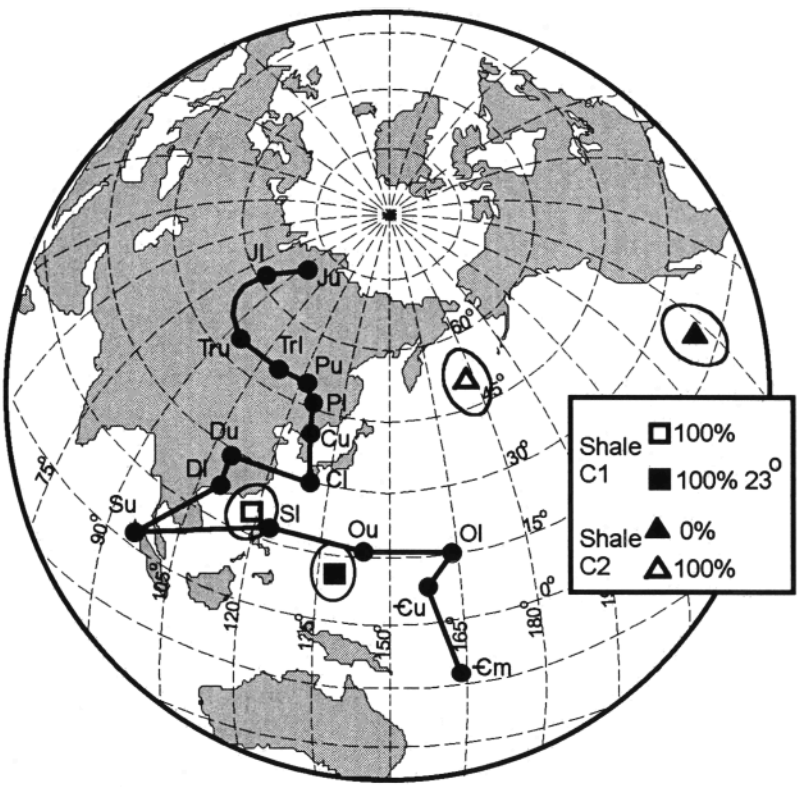


Fig. 12. The North American apparent polar wander path (solid circles connected with lines) from Cambrian to Jurassic using mean results with $Q \geq 3$ from *Van der Voo* [1990]. The shale C1 (square), shale C2 (triangle) poles are plotted along with their 95% confidence ellipses. The in situ C2 pole is plotted using a solid triangle; open symbols are in stratigraphic coordinates (see legend). The shale C1 pole falls on the lower Silurian portion of the APWP. The shale C2 poles lie far from any portion of the path. The shale C1 pole after 23° counterclockwise vertical axis rotation of the north limb of the Pennsylvania salient is plotted as a solid square. The rotated shale C1 pole falls on the upper Ordovician portion of the APWP, which would imply a (near-) primary origin for this direction.

previously mentioned, the lack of traditional paleomagnetic field tests precludes a statistical evaluation of our interpretation. We will therefore briefly present some alternative interpretations.

The first alternative is found by examining the in situ pole for the shale C1 direction, which falls directly on the lower Permian portion of the APWP (Figure 13). Use of this direction as the preferred characteristic direction for the shale calls for a postfolding, lower Permian (but normal polarity!) remagnetization or an upper Carboniferous synfolding remagnetization. This requires either that the rotations associated with oroclinal bending of the Pennsylvania salient did not affect the site or that such rotation was completed before the shales acquired their remanence. Also, because the shale C1 is of normal polarity, this remagnetization must have occurred either before the upper Carboniferous-lower Permian Kiaman superchron (but after folding), or else after the Kiaman superchron and that some later event remagnetized the slates but not the shales. Both of these possibilities are most unlikely.

Consideration of slate poles in alternative explanations first requires that these directions be considered paleomagnetically reliable. Examination of the slate poles finds that all, except for the in situ pencil slate pole, lie far from the APWP. The in situ pencil slate pole lies near the Jurassic portion of the path, which would call for a Jurassic remagnetization in the pencil slates, but not in the shales, and some "structural perturbance" of this Jurassic direction in the well-cleaved slates, which is not possible because the deformation is definitely older than the Jurassic. Correction for approximately 23° clockwise rotation of the Pennsylvania salient does, however, move the pencil slate poles (either in situ or tilt-corrected) nearer the Permo-Carboniferous segment of the APWP. This possibility is consistent with our preferred interpretation of the shale C1 direction, and it requires that the pencil slate direction is relatively unaffected by strain and is not a composite direction. Another, less likely possibility is that the well-cleaved slate directions are not affected by strain. Correction of the slate poles for rotation of the north limb of the Pennsylvania salient 23° (clockwise) would bring the well-cleaved (tilt-corrected) slate pole to the Silurian portion of the APWP. This possibility would require a Silurian remagnetization and neither strain modification nor magnetite dissolution and neocrystallization during Alleghenian deformation. This interpretation suffers from lack of evidence for Silurian remagnetizations in this area, and it is not consistent with the cleavage-parallel magnetite orientation found in these slates.

IMPLICATIONS FOR OTHER PALEOMAGNETIC STUDIES

Regardless of which of these interpretations is preferred, our study of the paleomagnetism of the shale to slate transition in the Martinsburg Formation shows that remagnetization of the slates is related to development of slaty cleavage. This remagnetization is probably late Paleozoic in age. Most importantly, the likely mechanism for this remagnetization was strain-induced dissolution and neocrystallization of magnetite. The exact cause of magnetite dissolution in these rocks is unknown but could be

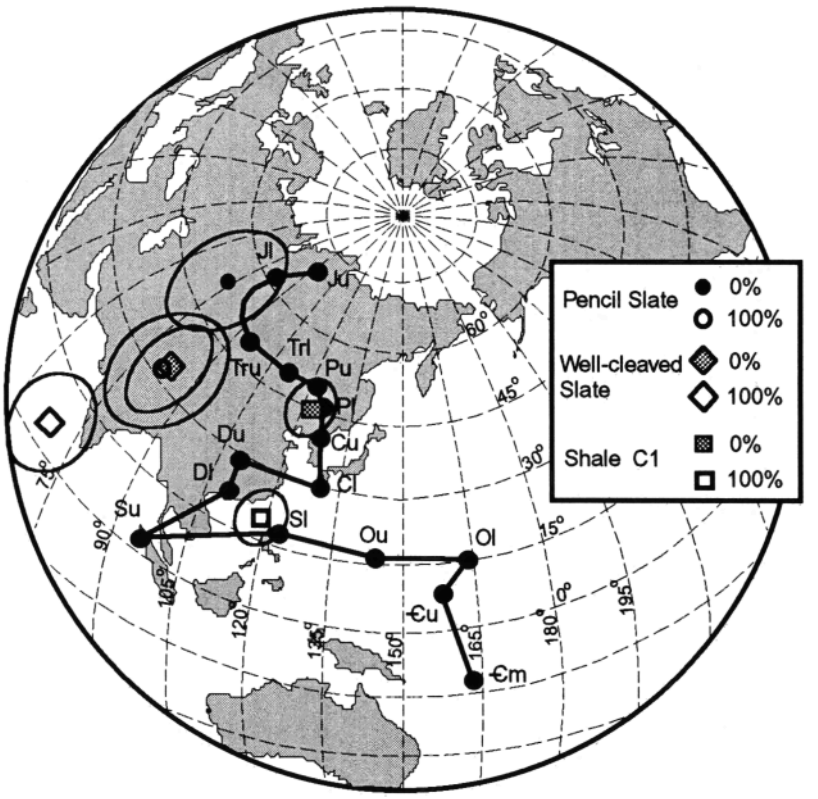


Fig. 13. North American APWP and poles for alternative interpretations of the Martinsburg Formation remanence. Pencil slate poles are denoted by circles, well-cleaved slate poles by diamonds, shale C1 poles by squares. In situ coordinate poles are denoted by solid symbols; stratigraphic coordinates by open symbols. See text for discussion.

related either to chemical changes induced by the dissolution of clay, mica and chlorite, or to pressure solution of magnetite, during cleavage development. The unknown quantitative relationship between dissolution and neocrystallization and strain precludes any attempt to correct the remanence for strain. This result shows that studies of cleaved sediments must consider the possibility of dissolution and neocrystallization of remanence carriers during deformation in addition to possible strain-induced rotation of the remanence carriers. In the Martinsburg Formation, inferred dissolution of magnetite occurred at temperatures below 300°C. This type of remagnetization can thus occur in other sediments which were folded under subgreenschist facies conditions. In light of these complications, paleomagnetic studies of folded and cleaved sediments should include detailed characterization of the remanence carriers, as well as their relationship with tectonic fabrics, through SEM, ARMA, and detailed rock magnetism.

The lack of a pervasive, reversed-polarity Kiaman superchron remagnetization in the shale has implications for the proposed mechanisms for this regional remagnetization event. The stratigraphic position of the Martinsburg Formation precludes a thermoviscous origin for the remagnetization in this part of the Appalachians, as the Martinsburg Formation is stratigraphically above the remagnetized Cambrian Allentown dolomite and stratigraphically below the remagnetized Devonian Onondaga limestone. The lack of a Kiaman remagnetization is consistent with an orogenic-fluid remagnetization mechanism [*Elmore and McCabe*, 1991], as the Martinsburg shale is relatively impermeable and therefore may have escaped the hot fluids that are thought to have been responsible for the remagnetization in the more permeable carbonate units.

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